rubber of bituminous components, ceramic materials or moulded plastic and are seated at the top to avoid the discharge of electrolyte. At the bottom of the container, there are four ribs, on two of them rest on the positive plate and the others support the negative plate.

The prism serves as the support for the plates, and at the same time protects them from short-circuit. The material which the battery containers are made should be resistant to sulphuric acid.

2. Plate

The plates of the lead acid cell is of diverse designs and they all consist some form of a grid which is made up of lead and the active material. The grid is essential for conducting the electric current and for distributing the current equally on the active material. If the current is not uniformly distributed, then the active material will loosen and fall out.

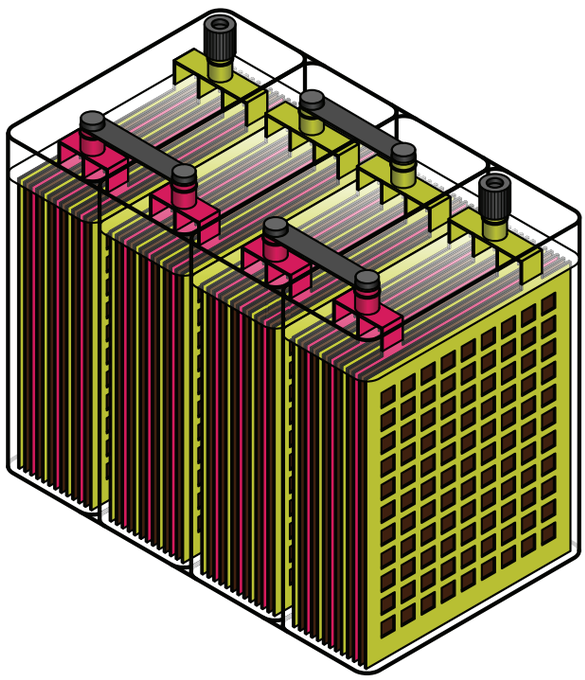


Fig 4.7 *Plate Arrangements of Lead-Acid Battery*

The grids are made up of an alloy of lead and antimony. The grid for the positive and negative plates are of the same design, (as shown in fig. 4.7) but the grids from the negative plates are made lighter because they are not as essential for the uniform conduction of the current.

The number of negative plates in a cell is always more than one number of positive plates in a cell, so that end plates at both the sides of the group remain negative.

3. Active Material

The material in a cell which takes active participation in a chemical reaction during charging or discharging is called the active material of the cell. The active element of the lead acid cells are

(a) Lead Peroxide (PbO_2)

It forms the positive active material. The PbO_2 is dark chocolate brown in colour.

b) Sponge Lead (Pb)

It forms the negative active material. It is grey in colour.

(c) Dilute Sulphuric Acid (H_2SO_4)

It is used as an electrolyte. It contains 31% of sulphuric acid.

4.Separators

The separators are thin sheets of non-conducting material made up of chemically treated leadwood, porous rubbers or mats of glass fibre and are placed between the positive and negative to insulate from each other. Separators are grooved vertically on one side and are smooth on the other side.

5. Battery Terminals

A battery has two terminals:–

Positive and Negative

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